CS 112

Introduction to Data Structures

WEEK 08

Algorithm Analysis and Big-Oh

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This Week

1. How long does each operation take?
2. Big-Oh notation
3. The operations, in Big-Oh **KEY**
4. $a \cdot f(n) + b$
5. Reading it off the code
6. Why it matters: electricity
7. `operator<<` for a container

Vector Subscript

```
v[i]    →    *(v.myArray + i * sizeof(Item))
```

> 3 operations, whatever *i* is and however big the vector is: **constant time**.

Vector Append

Room left

```
myArray[mySize] = it;
++mySize;
```

a few operations: **constant**

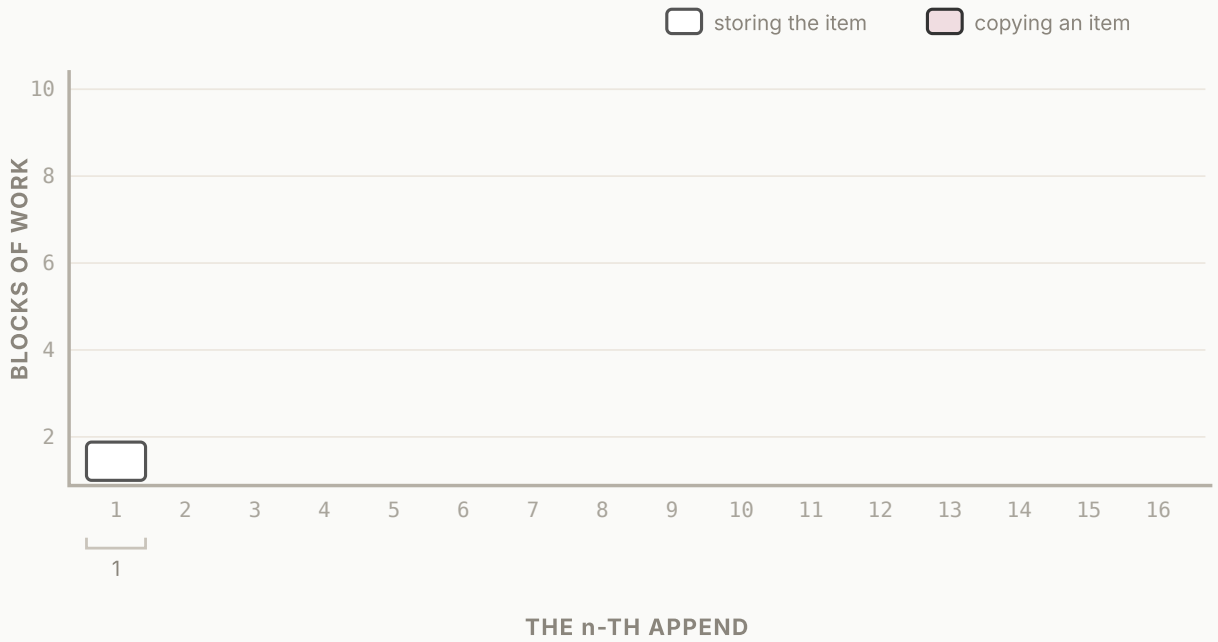
Full

```
new Item[2 * mySize]
copy n items
delete [] old
```

about 3*n* + 5: **linear**

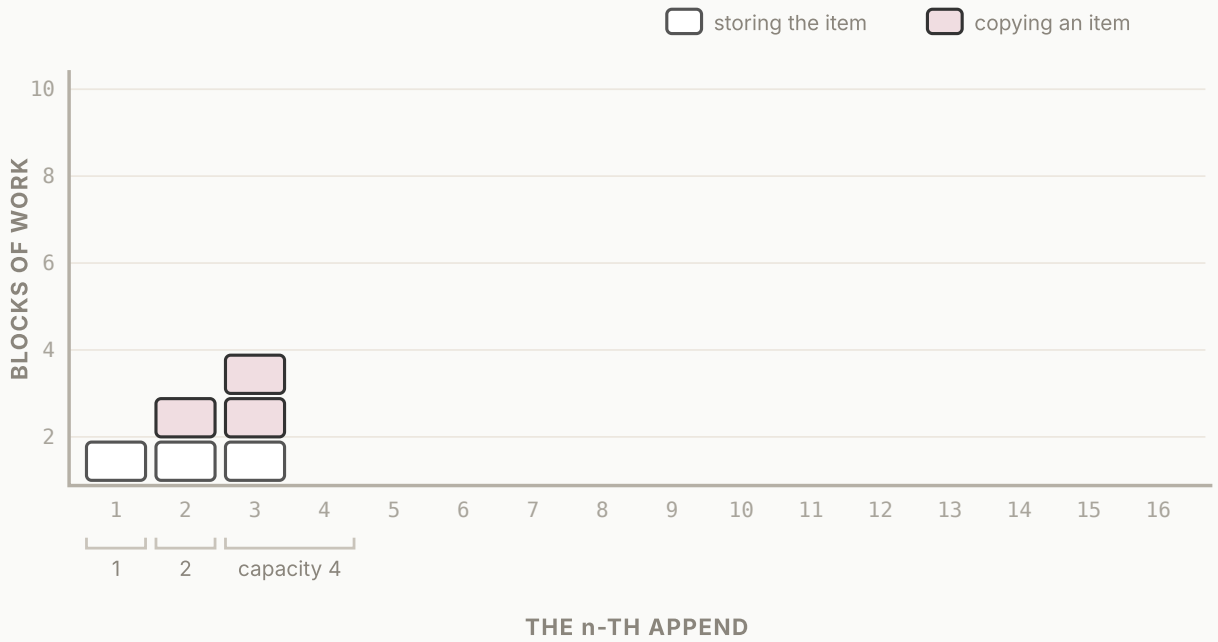
Vector Append, on Average

1. Append 1: there is room, so it just stores the item.



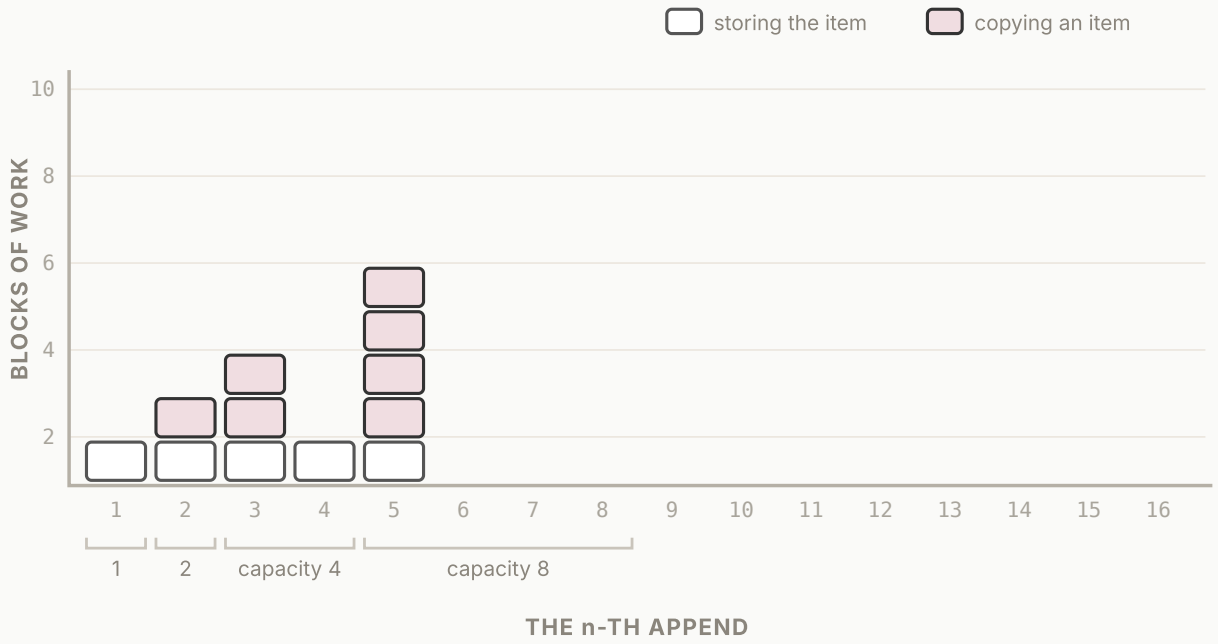
Vector Append, on Average

2 Append 3: the array is full, so it copies 2 items first.



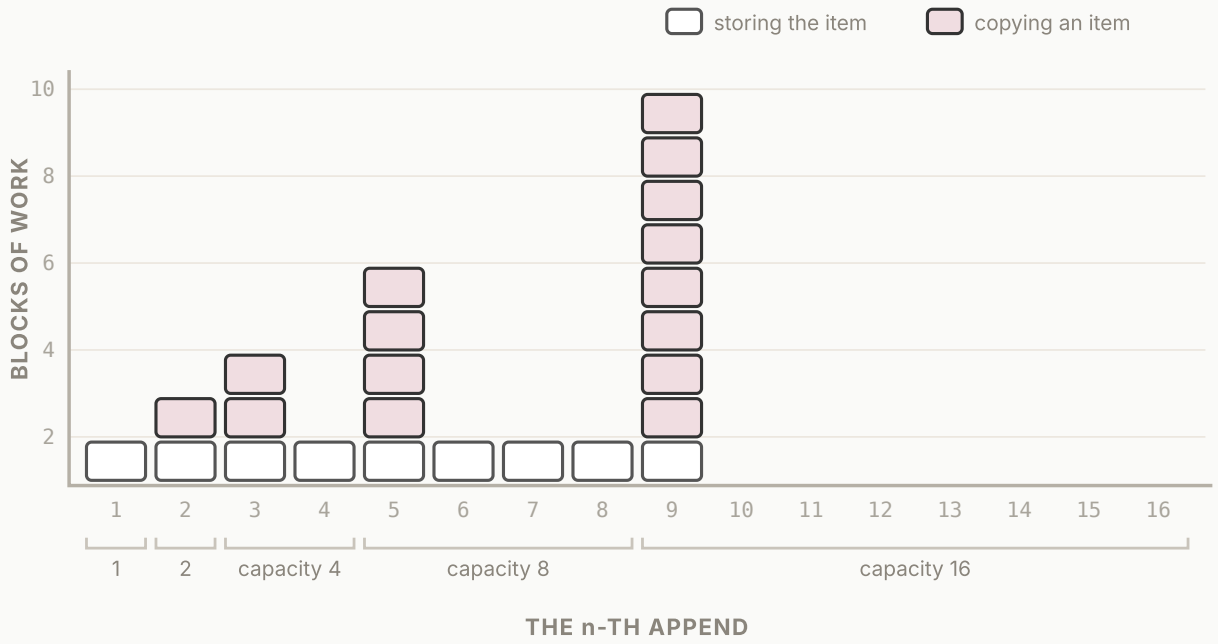
Vector Append, on Average

3 Append 5: the array is full, so it copies 4 items first.



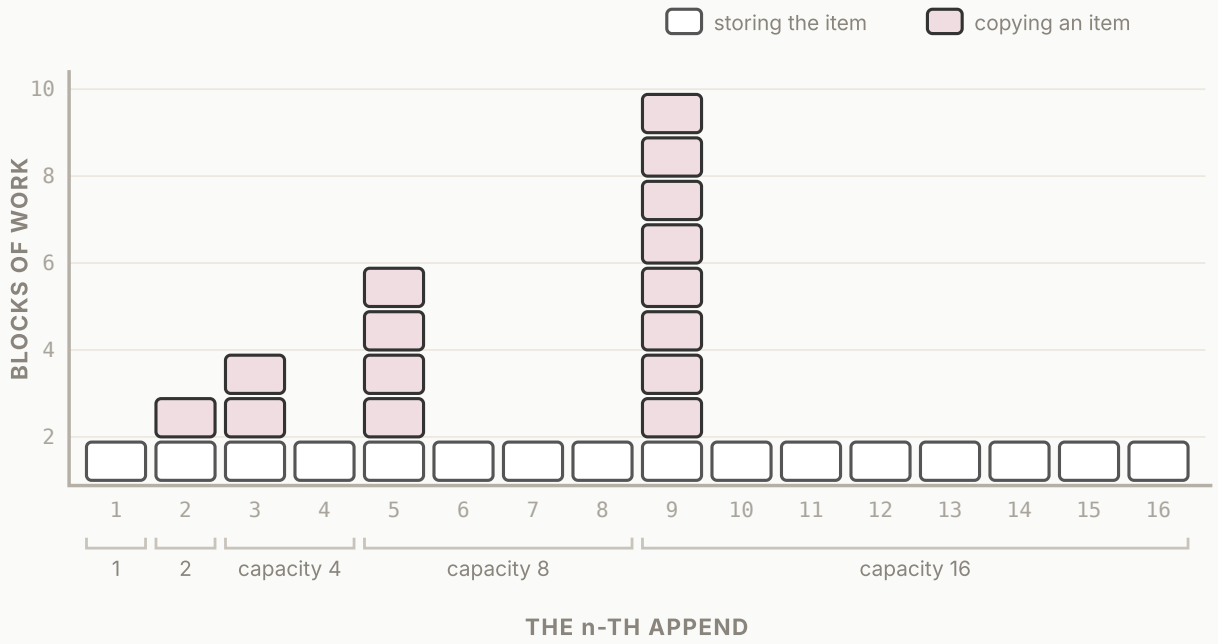
Vector Append, on Average

4 Append 9: the array is full, so it copies 8 items first.



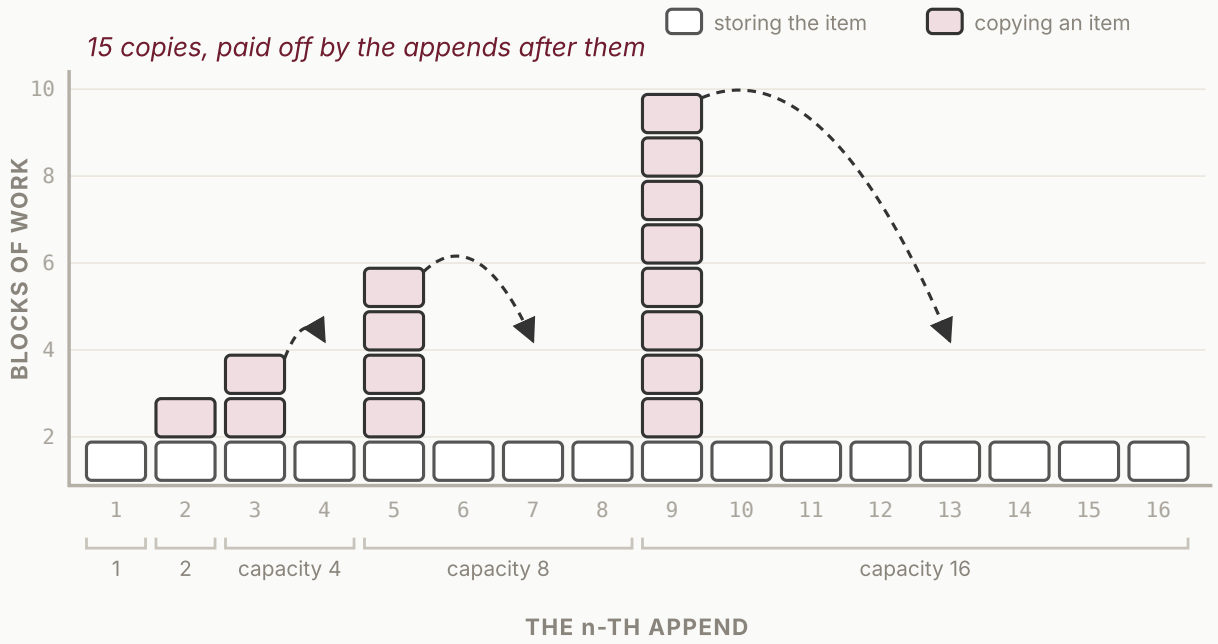
Vector Append, on Average

5 Append 16: there is room, so it just stores the item.



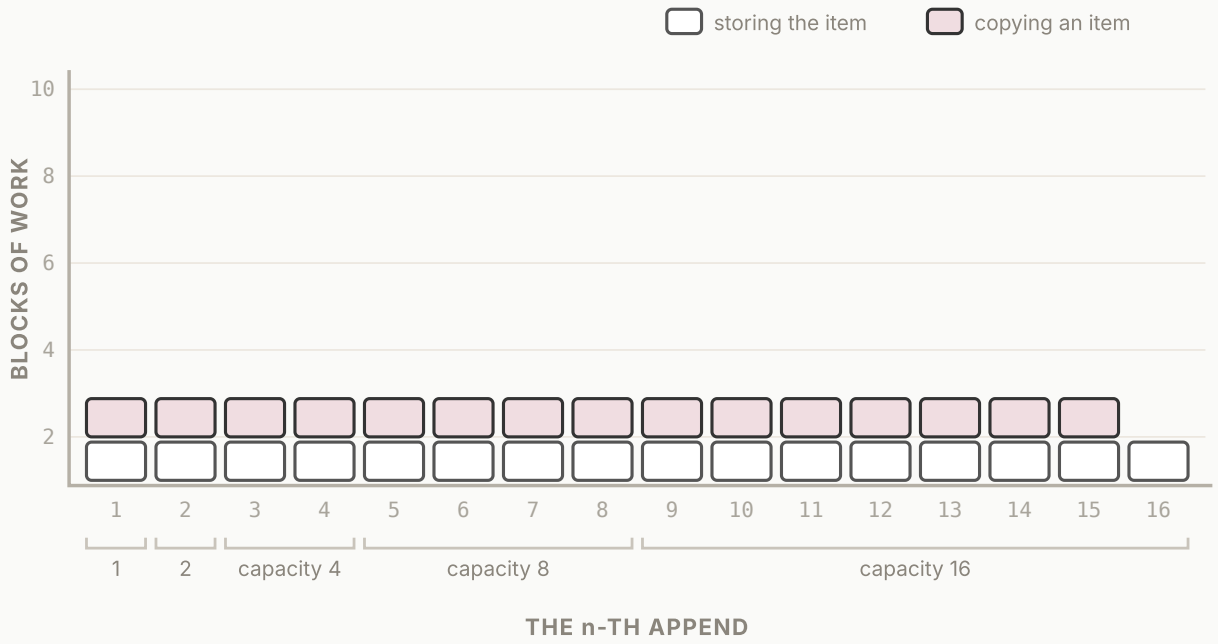
Vector Append, on Average

6 Each spike is paid for by the appends that follow it.



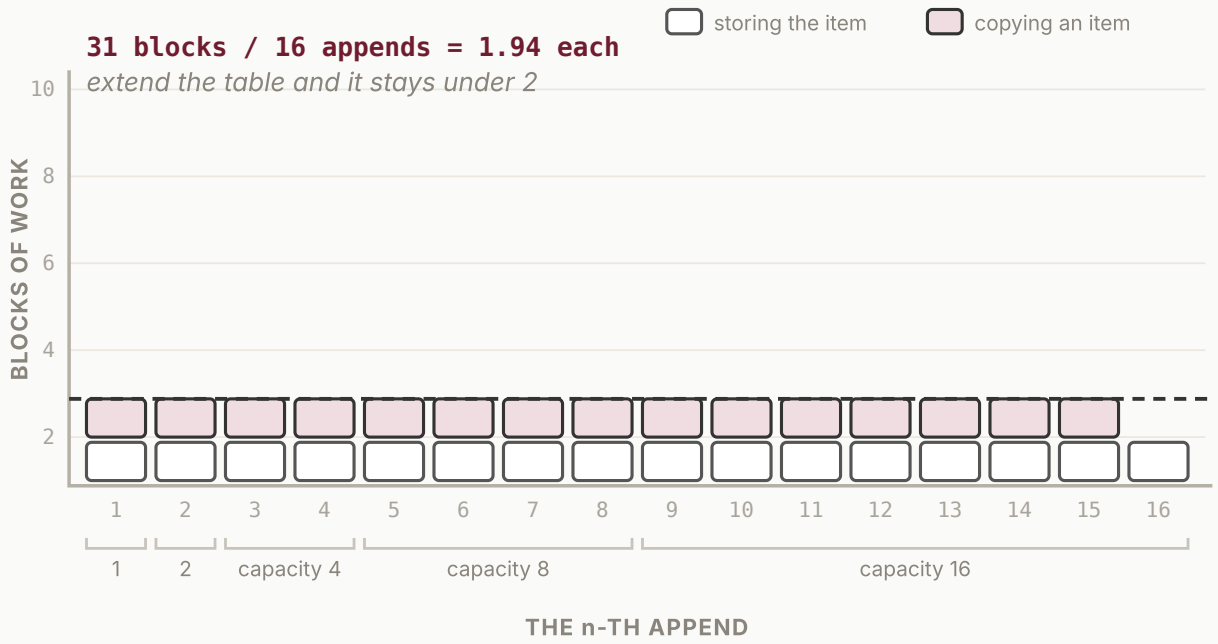
Vector Append, on Average

7 The same blocks, laid flat. Nothing was added or removed.



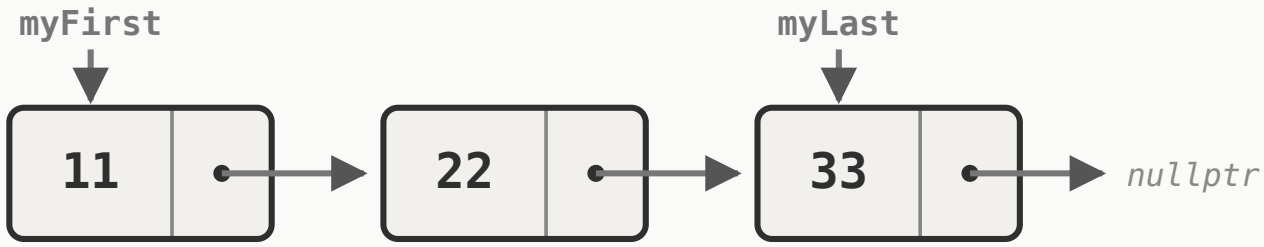
Vector Append, on Average

8 31 blocks over 16 appends, and the average does not grow with n.



Amortized constant time.

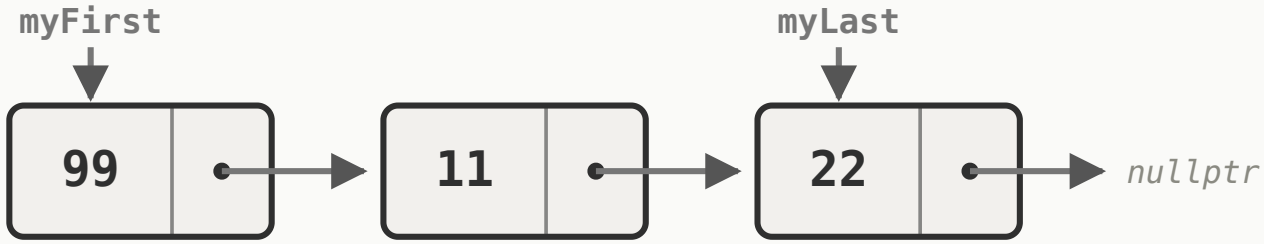
List Append




```
Node* n = new Node(it, nullptr);
if (mySize == 0) myFirst = n;
else             myLast->myNext = n;
myLast = n;
++mySize;
```

> About 6 operations, empty list or not: **constant**.

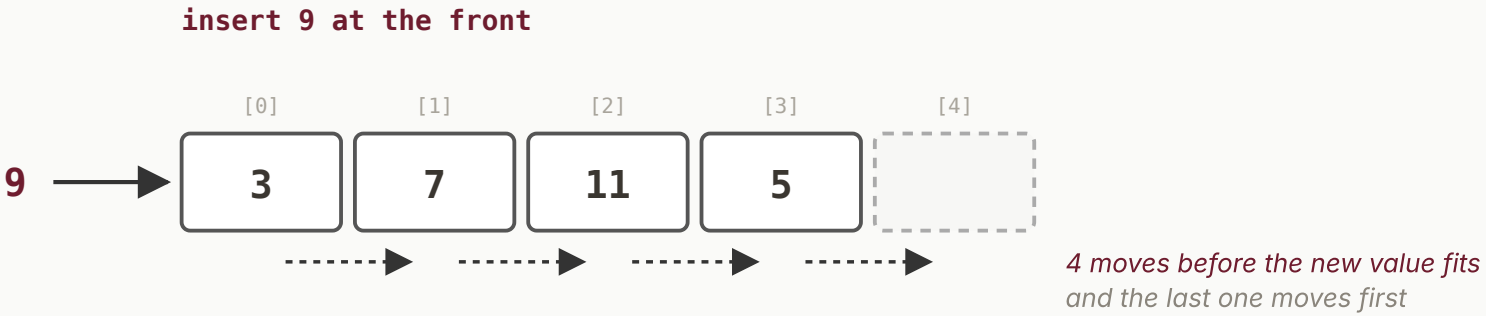
List Prepend




```
myFirst = new Node(it, myFirst);
if (mySize == 0) myLast = myFirst;
++mySize;
```

> About 5 operations, empty list or not: **constant**.

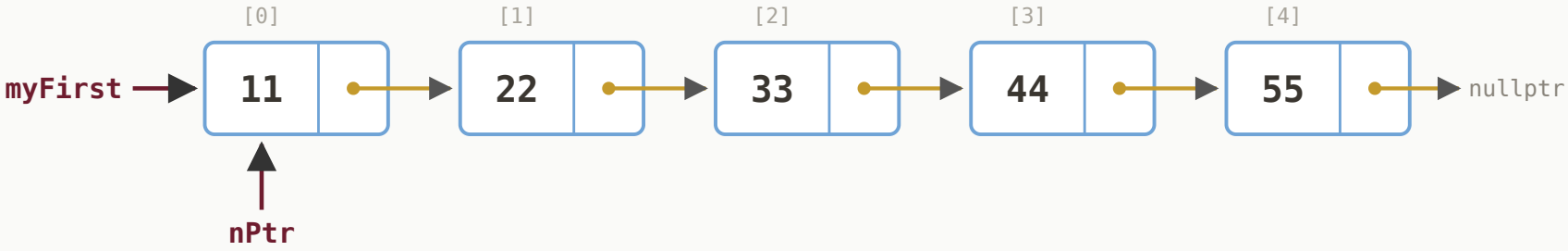
Vector Prepend



Every item shifts first: **linear**. So `vector` has no `push_front`.

List Subscript

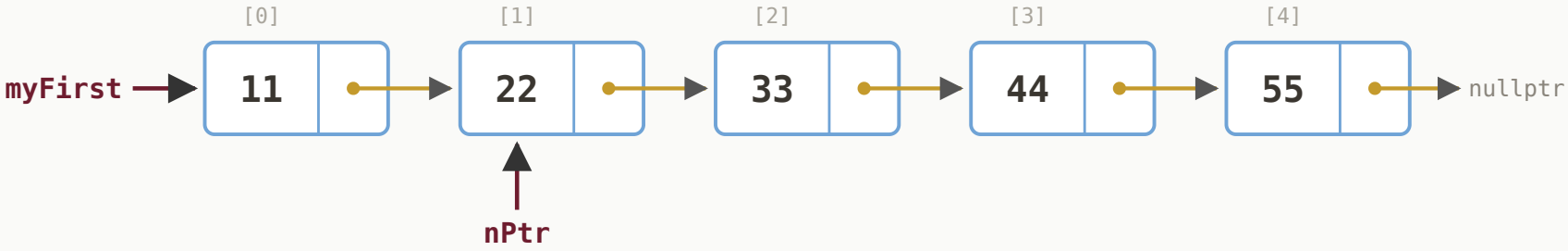
1. l[3] starts where every walk starts: nPtr = myFirst.




```
Item& List::getIndex(unsigned ix) {
  Node* nPtr = myFirst;
  for (unsigned i = 0; i < ix; i++)
    nPtr = nPtr->myNext;
  return nPtr->myItem;
}
```

List Subscript

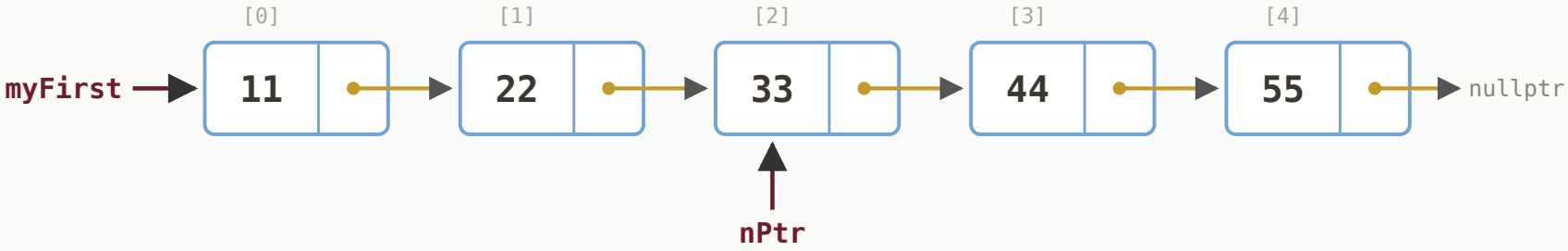
2 One pass of the loop, one hop: nPtr = nPtr→myNext.




```
Item& List::getIndex(unsigned ix) {
  Node* nPtr = myFirst;
  for (unsigned i = 0; i < ix; i++)
    nPtr = nPtr->myNext;
  return nPtr->myItem;
}
```

List Subscript

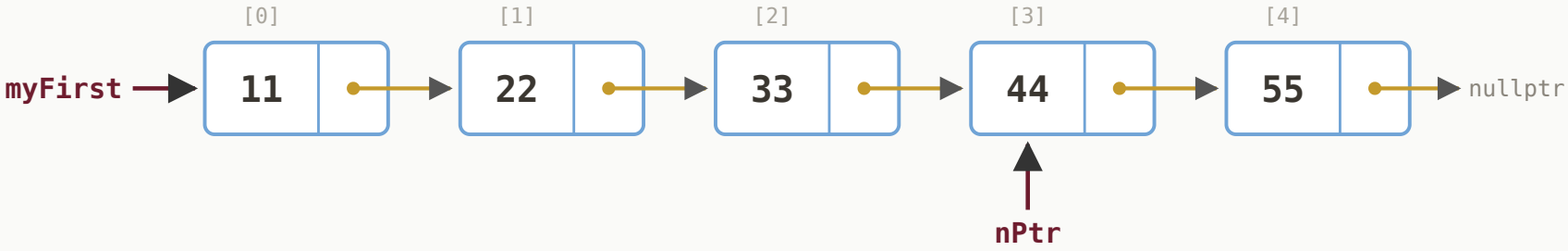
3. Another hop. There is no way to jump ahead.




```
Item& List::getIndex(unsigned ix) {
  Node* nPtr = myFirst;
  for (unsigned i = 0; i < ix; i++)
    nPtr = nPtr->myNext;
  return nPtr->myItem;
}
```

List Subscript

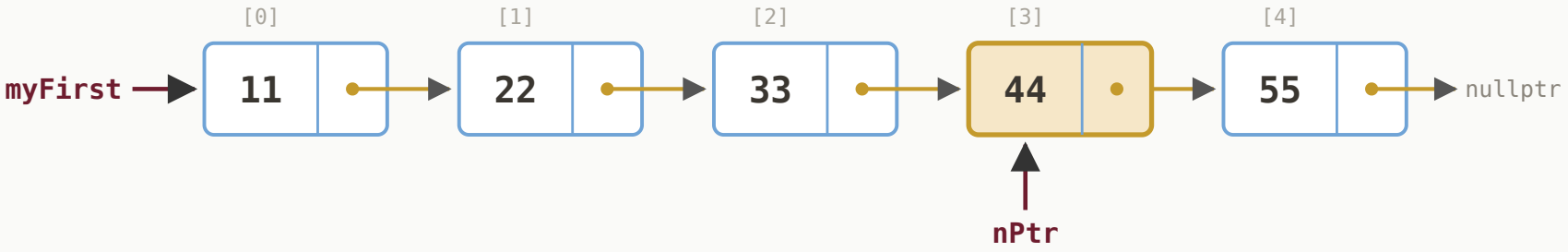
4 The third hop reaches item 3, and i == 3 ends the loop.




```
Item& List::getIndex(unsigned ix) {
  Node* nPtr = myFirst;
  for (unsigned i = 0; i < ix; i++)
    nPtr = nPtr->myNext;
  return nPtr->myItem;
}
```

List Subscript

5 Three hops for index 3. The last item takes n - 1, so on average the loop runs about n/2 times: linear.




```
Item& List::getIndex(unsigned ix) {
  Node* nPtr = myFirst;
  for (unsigned i = 0; i < ix; i++)
    nPtr = nPtr->myNext;
  return nPtr->myItem;
}
```

> About *n*/2 hops on average: **linear**. So STL's `list` has no `[]`.

TALK TO YOUR NEIGHBOR · TTYN

Which of these is linear?

A. `v[500]` on a vector of 1000

B. `push_front` on a list of 1000

C. inserting at the front of a vector of 1000

D. `push_back` on a list of 1000

Big-Oh Notation

Complexity	Big-Oh	
constant	$O(1)$	polynomial
logarithmic	$O(\lg n)$	
linear	$O(n)$	
quadratic	$O(n^2)$	
cubic	$O(n^3)$	
exponential	$O(2^n)$	**non-polynomial**
factorial	$O(n!)$	

> Better at the top, worse at the bottom.

The Operations, in Big-Oh

Operation	Complexity	Big-Oh
vector subscript	constant	O(1)
vector append	amortized constant	**O(1*)**
vector prepend	linear	O(n)
list append	constant	O(1)
list prepend	constant	O(1)
list subscript	linear	O(n)
shallow copy	constant	O(1)
deep copy, assignment	linear	O(n)
input or output	linear	O(n)

> Need an operation the container doesn't have? Choose a different container.

What O(f(n)) Means

$$\text{exact cost} = a \cdot f(n) + b$$

> As *n* grows, f(n) dominates. Nobody cares about *a* and *b*.

At a Million Items

Operation	Exact cost	At n = 1,000,000	Big-Oh
vector subscript	3	3	O(1)
list append	5	5	O(1)
vector prepend	$3n + 2$	3,000,002	O(n)
list subscript	$0.5n + 2$	500,002	O(n)

> 3 and 5 are both "about 1" next to a million. 3,000,002 and 500,002 are both "about a million".

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An algorithm takes `4n + 200` steps. Its complexity is...

A. O(4n)

B. O(n)

C. O(n + 200)

D. O(200)

Reading It Off the Code

```
for (int i = 0; i < n; ++i) {
  total += a[i];
}
```

one loop over n: **O(n)**

```
for (int i = 0; i < n; ++i) {
  for (int j = 0; j < n; ++j) {
    if (a[i] == a[j]) ...
  }
}
```

a loop inside a loop: **$O(n^2)$**

> Loops one after another add, keep the biggest. Nested loops multiply.

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What is the complexity of this loop?

```
for (int i = 1; i < n; i = i * 2) {
    cout << i << endl;
}
```

A. O(n)

B. $O(n^2)$

C. O(lg n)

D. O(n lg n)

Searching a Million Items

Search	Big-Oh	Comparisons
linear search, any vector	O(n)	up to 1,000,000
binary search, sorted vector	O(lg n)	about 20

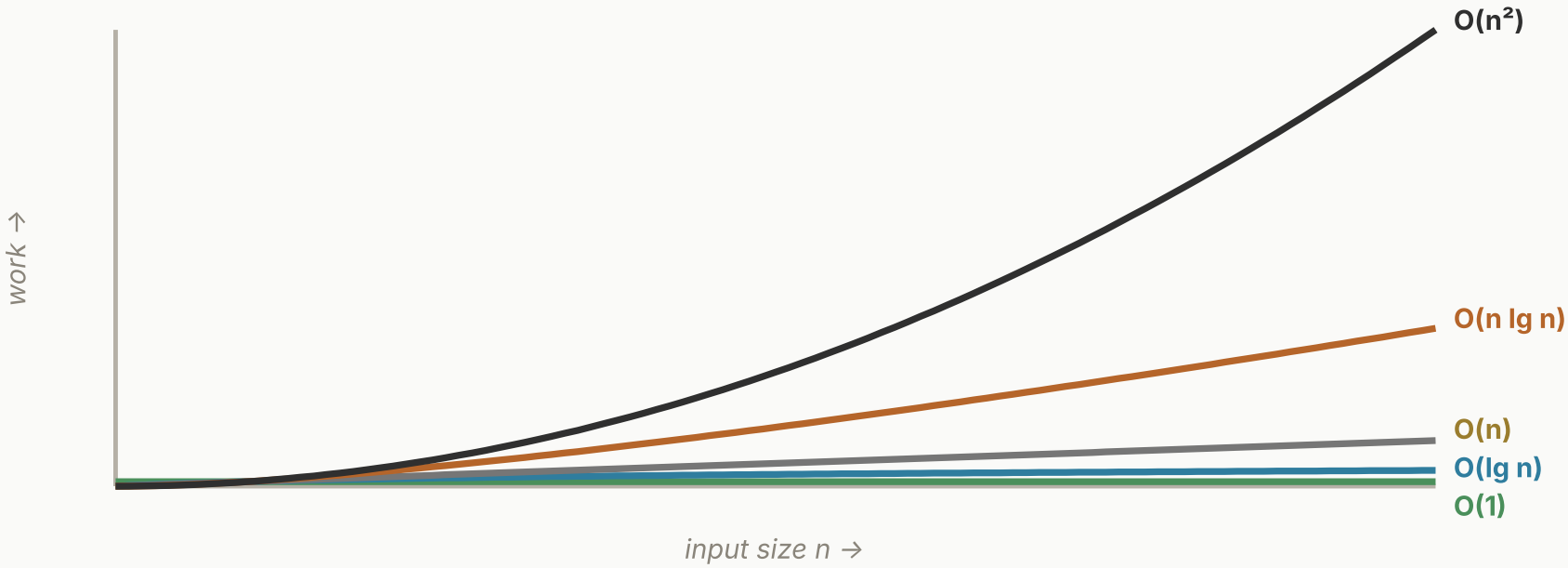
> Computers run on electricity. Fewer comparisons, less electricity, less carbon.

Sorting a Million Items

Sort	Big-Oh	Steps, roughly
bubble sort	$O(n^2)$	1,000,000,000,000
quicksort	O(n lg n)	20,000,000

> Big-Oh lets us estimate an algorithm's **carbon footprint**.

Visualizing Big-Oh



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A is $O(n^2)$, B is $O(n \lg n)$. Which should you use?

A. Always B

B. Always A

C. B for large n; for small n, A may be faster

D. Whichever is easier to write

operator<<

```
cout << aList << endl;

// is really
operator<<( operator<<(cout, aList), endl );
```

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What should `operator<<(ostream& out, const List& list)` return?

A. nothing, it is `void`

B. `list`

C. `out`

D. `true`

operator<<

```
ostream& operator<<(ostream &out, const List &list) {
    list.writeTo(out);
    return out;
}
```

> `out` is not `const`: writing to it changes it.

Week 08 Recap

- Big-Oh describes **how work grows with n**, not seconds on a clock
- exact cost = $a \cdot f(n) + b$; keep f(n)
- Vector append is `O(1*)`, amortized constant
- Nested loops multiply; halving the problem gives `lg n`
- Better algorithms use less electricity

Next: a09 →

This deck stands on earlier CS112 materials by
Joel Adams and **Victor Norman** · adapted and extended by **Eric Araújo**